Project Executable Files

Date	29 June 2026
Team ID	
Project Name	OptiCrop : Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	YES
2	README / Setup Guide (instructions to run the project)	YES
3	requirements.txt / package.json / dependency file	YES
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	YES
6	Deployed application URL (if hosted)	NA
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
AI-ML-and-GEN-AI-Track-Project-main/
|
├── 1. Brainstorming & Ideation/
|   ├── Brainstorming & Idea Prioritization.pdf
|   ├── Define Problem Statements.pdf
|   └── Empathy Map.pdf
|
├── 2. Requirement Analysis/
|   ├── Customer Journey Map.pdf
|   ├── Data Flow Diagram.pdf
|   ├── Solution Requirements.pdf
|   └── Technology Stack.pdf
|
├── 3. Project Design Phase/
|   ├── Problem-Solution Fit.pdf
|   ├── Proposed Solution.pdf
|   └── Solution Architecture.pdf
|
├── 4. Project Planning Phase/
|   └── Project Planning.pdf
|
├── 5. Project Development Phase/
|   └── code/
|       ├── app.py
|       ├── train.py
|       ├── notebook.ipynb
|       ├── requirements.txt
|       ├── .gitignore
|       |
          ├── README.md
|       ├── data/
```

```
│      │    ├── Crop_recommendation.csv
│      │    └── crop_info.py
│      │
│      ├── models/
│      │    ├── crop_model.pkl
│      │    ├── scaler.pkl
│      │    └── model_info.pkl
│      │
│      ├── templates/
│      │    └── index.html
│      │
│      └── static/
│          ├── css/
│          │    └── style.css
│          └── js/
│              └── script.js
│
├── 6. Project Testing/
│    └── Performance Testing.pdf
│
├── 7. Project Documentation/
│    ├── Project Executable Files.pdf
│    └── Sample Project Documentation.pdf
│
└── 8. Project Demonstration/
    ├── Communication.pdf
    ├── Demonstration of Proposed Features.pdf
    ├── Project Demo Planning.pdf
    ├── Scalability & Future Plan.pdf
    └── Team Involvement in Demonstration.pdf
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	NA
Login Credentials (Demo)	Not Applicable
Platform / Hosting Provider	NA
Repository Link	GitHub Repo Link
Demo Video Link	Demo Video Link

Step 4: Run Instructions

Pre-requisites

- Python **3.11+**
- Git
- Visual Studio Code (recommended)
- Internet connection (for Bootstrap CDN)

Steps

Step 1

Clone the repository **git clone <repository-url>**

Step 2

Navigate to the project folder **cd OptiCrop**

Step 3

Create Virtual Environment **python -m venv venv**

Step 4

Activate Virtual Environment

Windows **venv\Scripts\activate**

Linux/Mac **source venv/bin/activate**

Step 5

Install Dependencies **pip install -r requirements.txt**

Step 6

Train the Machine Learning Model **python train.py**

Step 7

Run Flask Application **python app.py**

Step 8

Open Browser http://127.0.0.1:5000

Step 5: Known Issues/ Limitations

S.No	Known Issues / Limitations	Workaround/Status
1	Application currently runs only on the local Flask development server	Can be deployed to Render, Railway, or Azure in future
2	Weather API and fertilizer recommendation are not implemented	Future enhancement
3	Crop images are not yet integrated	Planned for the next version